WORKSPACE · CODING_AND_RUNNING

ocl-raghav

23 files

GPU $ PROGRESSION (CROSS-BRANCH)

ONE-LINE (WORKSPACES.XML)

iter6 scientist: iter5 CRITICAL audit accepted. R26 claim-bearing count corrected to 16/30 triple-strict (not aggregate 20/30); R25 found 30 hard cells under unchanged 0.88 gate. Next coder round R27-R30: raw evidence sync, boundary/gamma-ceiling rescue, calibration-only k/q selection, coverage robustness. phase=coding_and_running.

STATE

PHASE	coding_and_running
VERSION	0
ITERATION	6
ROUTE	selected-risk-calibration
BRANCH	route/selected-risk-calibration
GPU $	$222.61

STATE.MD

项目名：ocl-raghav

> 一句话：研究 conformal learned constraints 在 optimizer 返回点上为什么会失效，并用 selected-risk calibration / scale-aware calibration 把可行性保证从随机测试点转移到 solver 真正返回的决策点。

1. 背景

1.1 领域常识

OCL 学一个未知约束 `h(x)<=0` 的 surrogate `h_hat(x)`，再让优化器在 learned feasible region 里找最优决策。风险是 optimizer 会主动寻找 surrogate 的 false positive，而随机点上看起来安全的 conformal margin 不一定保护 optimizer 返回点。

Split conformal 的标准保证针对 exchangeable random point。这里真正交付的是 `x*(c)=argmin c(x)` under `h_hat(x)+q_hat<=0`； `x*` 是被 solver 和目标函数选出来的点，不是随机测试点。

本路线的核心问题是：随机点 coverage 接近 90% 时，selected solution 是否仍可能远低于 90% 可行？如果会，必须在 objective-induced selected distribution 上校准，而不是只在 random x 上校准。

1.2 核心概念

概念	一句话解释
learned constraint	用数据学出的约束函数 `h_hat(x)`，代替未知真实约束 `h(x)`
split conformal margin	在随机 calibration set 上取 one-sided residual 分位数 `q_hat`
optimizer-selected feasibility	optimizer 返回的 `x*` 在真实约束 `h(x*)<=0` 下是否可行
selected-risk calibration (SRC/OSRC)	用 held-out objectives 产生的 optimizer 返回点校准额外 margin
scale-aware SRC	用局部 residual scale `s_hat(x)` 替代全局 scalar margin
ACI	在线逐个 objective 更新 margin，使长期 selected violation 接近目标 `alpha`
alpha-tightening	直接把随机点 conformal 的 alpha 设得更小；这是必须正面对比的强 baseline

1.3 研究锚点

项	当前定义
Bottom-line problem	OCL 的可行性证书必须覆盖 optimizer 返回的决策点，而不是只覆盖随机测试点
Primary claim	随机点 conformal coverage 不足以保证 selected feasibility；selected-risk calibration 能以有限样本方式控制未来 objective 分布下的 selected violation
Supporting claim	ACI 是在线场景的实用对应方法；符号约束是 solver-friendly base model，但不是主保证
Anti-claim	如果 alpha-tightening、dose response、trust region 或 C-MICL-style baseline 以很小 objective cost 达到 selected feasibility >=90%，主方法就没有 load-bearing 优势
Non-goals	不声称发明 conformal prediction、LLM-SR、symbolic regression、global nonlinear optimization，也不声称 per-instance conditional coverage
Minimum convincing evidence	至少两个有效 hard selected-gap cells；SRC/variant 在这些 cell 上控制 selected violation 且不被强 baseline frontier 支配；cost 用 matched feasible subset 计算

2. 实验全景

2.1 实验流程

1. 生成 oracle data：train / random-cal / random-test / objective-cal / objective-test。
2. 训练 surrogate `h_hat`：symbolic pinball、MLP、或 C-MICL-style baseline。
3. 用随机点 split conformal 得到 `q_hat`，先检查 random coverage。
4. 对很多 objective 解 static OCL，oracle 检查 selected `x*`。
5. 用 additive SRC、scale-aware SRC 或 ACI 在 selected distribution 上校准。
6. 同步强 baseline frontier：alpha-tightening、dose response、trust region、C-MICL-style controls。

2.2 核心指标

指标	含义	当前使用方式
random coverage	随机测试点是否被 conformal upper bound 覆盖	低于 0.88 的 surrogate cell 不能支持 selected-gap claim
selected feasibility	optimizer 返回点是否真实可行	主问题指标
selection gap	random coverage - selected feasibility	static failure 的核心数字
selected violation	`1 - selected feasibility`	SRC/ACI/controls 的安全指标，目标 alpha=0.10
matched objective degradation	安全 margin 的优化代价	只在 both-method finite-and-oracle-feasible objective subset 上算
solver feasible rate	downstream solver 是否仍能找到可行点	trust/dose controls 的必要解释变量

2.3 实验矩阵

实验	研究问题	状态	核心结果
R1-R2	methane/Williams static gap	collected	methane NULL；Williams/sym gap +0.079 CI[0.073,0.091]
R6-R8	Williams additive SRC / ACI / controls	collected	additive fails seed1; ACI support; α-tight α=0.01 violation 0.108 at +1.07%
R9-R13	portfolio3d/mlp gap, SRC, controls, ACI	collected	gap +17pp real, but α-tight/dose close cheaply; ACI burst high
R14-R15	Williams/sym scale-aware SRC + bias audit	collected	kNN violation 0.034 at +8.1%; failures localize to T=500 boundary
R16	portfolio3d/sym coverage rescue	collected	cov mean 0.907, min 0.885; gap +0.090, controls then needed
R17	8-benchmark baseline-gated hard-cell screen	collected	2304 cells; 0/48 unique cells pass all 4 gates
R18	scale-aware SRC on R17 hard cells	blocked by R17	no hard cells existed
R19	Williams/sym kNN scale frontier	collected	best k=10/q=0.9 violation 0.029 at +4.14%; 17/20 configs cost >10%
R20	Williams/sym MLP gamma extension	collected	gamma=10 genuine; MLP scale NEGATIVE at +26.5% cost
R21	portfolio3d/sym robustness+controls+SRC	collected	coverage/gap robust, but α-tight α=0.005 dominates at +2.68%
R22	provenance repair	collected	R7/R12 manifests repaired, validation all_ok=true

实验	研究问题	状态	核心结果
R23-R26	audit sync, locked coverage repair, parameterized stress suite, scale-aware on discovered hard cells	collected	R25 finds 30/216 hard cells; R26 repairs 16/30 triple-strict

3. 算法与代码

3.1 算法本质

Static conformal OCL：训练 `h_hat`，在随机 calibration 点上取 `q_hat`，然后解 `min c(x)` subject to `h_hat(x)+q_hat<=0`。它验证的是 random coverage，不验证 optimizer 返回点的分布。

Additive SRC/OSRC：对 candidate `lambda` 和 calibration objectives 解 OCL，oracle 检查 selected violations，选 UCB selected-risk <= alpha 的最小 `lambda`。R6/R11 说明 scalar margin 能控安全，但在 Williams/sym 上不稳，在 portfolio3d/mlp 上没有相对强 baseline 的优势。

Scale-aware SRC：部署 `h_hat(x)+q_hat+gamma*s_hat(x)<=0`，其中 `s_hat(x)` 是局部 residual scale。R26 在 R25 hard cells 上给出 claim-valid 正信号：16/30 cells 同时满足 all-seeds-pass、violation<=0.10、cost<=5%。但失败集中在 boundary family 且 gamma 接近上限，说明当前 kNN scale 估计器还不够稳。

ACI：objective 顺序到来时，每次 solver 返回 `x_t` 后查询 oracle；若 infeasible 就增大 margin，若 feasible 就按步长回落。R7/R13 支持 long-run control，但 burst 指标必须报告。

3.2 代码地图

想知道...	文件	函数/脚本
conformal margin	`LLM-SR/llmsr/conformal.py`	`calibrate()`, `conformal_quantile()`
OCL solver	`LLM-SR/llmsr/ocl_solver.py`	`solve()`, `solve_constrained()`
benchmark oracle	`LLM-SR/benchmarks/`	`methane_reactor.py`, `williams_otto.py`, `portfolio3d.py`
static selected gap	`LLM-SR/scripts/run_selected_feasibility.py`	R1/R2/R9/R10/R16-style runner
additive SRC	`LLM-SR/scripts/run_selected_risk_calibration.py`	R3/R6/R11 runner
scale-aware SRC	`LLM-SR/scripts/run_scale_aware_src.py`	R14/R19 runner
ACI recovery	`LLM-SR/scripts/run_aci_selected_feasibility.py`	R4/R7/R13 runner
strong controls	`LLM-SR/scripts/run_selected_controls.py`	R5/R8/R12/R17 controls

3.3 计算量

当前主线是 CPU/SLSQP + small MLP，不需要 GPU。node05 适合 objective sweeps 与大批量 cell/decision 同步：2026-07-07 23:51 EDT health 显示 1h CPU util 3%、12h 12%、10.6T free、无 GPU。node05 现在有 screen/tmux，但本项目仍沿用 direct `nohup + job.pid` + logs + manifest receipt；用 Python 3.12 env 避免 scipy 1.12 在 py3.13 上源码构建。

4. 当前结果

4.1 已完成证据

Evidence	关键数字	科学判断
Static selected-gap evidence	Williams/sym +7.9pp；portfolio3d/mlp +17.0pp；portfolio3d/sym +9.0pp；Williams/MLP repaired +3.3pp	selected failure mode real
R17 broad hard-cell screen	2304 cells；0/48 unique cells pass all 4 gates	generic benchmarks insufficient; §5 localized-stress route needed
R24 locked Williams repair	random_cov_min 0.881；static violation 0.166；gap 0.058；best kNN violation 0.043 at +3.61%	valid by violation, but gap below 0.08 and kNN Pareto is isolated
R25 parameterized stress suite	10368 cells；30/216 hard cells pass all gates；26/30 have coverage 0.881-0.889；0 corner cells pass	§5 mechanism works, but many cells sit near coverage gate
R26 scale-aware on R25 cells	1800/1800 tasks；16/30 triple-strict；20/30 all_seeds_pass；24/30 viol<=0.10；best cost +0.46%	main method signal now positive, with exact 16/30 count
R26 failure pattern	10 all-seeds failures; 9/10 boundary/symbolic; gamma_mean >=7.17	structural kNN limitation, not noise

4.2 Claim Matrix

Claim	当前证据	状态	Minimum evidence still needed
Random conformal coverage can fail on optimizer-selected points	R25 has 30 hard cells passing coverage/gap/violation/no-cheap-baseline gates	supported	sync raw cells/decisions locally for audit
Selected-risk calibration is necessary relative to simple baselines	R25 found 30 hard cells with no cheap baseline; R26 kNN SRC succeeds on 16/30 triple-strict; 20/30 all_seeds_pass; 24/30 viol<=0.10	supported	validate remote-only evidence and pre-register k/q selection
Scale-aware SRC can repair localized selected-risk failure	R26 triple-strict cells include 7 interior/MLP, 5 interior/sym, 4 boundary/sym; best cost +0.46%	supported	repair or diagnose 10 boundary/gamma-ceiling failures
Additive SRC is sufficient	R6 fails; R11 dominated/tied	contradicted	do not use additive SRC as final method evidence
ACI is useful online support	R7/R13 long-run control, burst caveat	support only	report burst and do not use as primary evidence

4.3 Scientific Interpretation

Observation: R25 turned R17's zero-hard-cell result into 30 valid hard cells by explicitly parameterizing localized residual pockets and boundary-risk objectives.
Interpretation: the selected-risk failure is not generic; it appears when residual error is spatially localized and objectives concentrate on the risky boundary. This supports §5's mechanism without relaxing the 0.88 coverage gate.
Alternative explanations: 26/30 hard cells barely clear 0.88 coverage, so some cells may be artifacts of a coverage cliff rather than robust examples.
Implication: R30 must re-test coverage and gate stability with independent random-test samples before broadening the main evidence table.
Next experiment: R30 coverage-cliff robustness.

Observation: R26 repairs 16/30 hard cells under the strict intersection of all_seeds_pass, violation<=0.10, and cost<=5%.
Interpretation: kNN scale-aware SRC is now real method evidence and beats the A1 minimum by a wide margin, but the report's 20/30 aggregate was not the triple-strict count.
Alternative explanations: post-hoc k/q selection may inflate the signal; raw R25/R26 decisions are still remote-only.
Implication: R27 must sync/validate raw evidence, and R29 must replace post-hoc best with a pre-registered k/q selection rule.
Next experiment: R27 sync validation and R29 nested k/q selection.

Observation: all 10 R26 all-seeds failures have gamma_mean >=7.17, and 9/10 are boundary/symbolic cells.
Interpretation: steep boundary residuals break the current kNN scale estimator; simply extending the same grid may raise cost without solving local geometry.
Alternative explanations: gamma ceiling could be an artificial grid cap, or boundary failures may require anisotropic/distance-to-boundary scale features.
Implication: R28 should test stronger boundary-aware scale estimators and an extended gamma diagnostic on the failing cells.
Next experiment: R28 boundary failure rescue.

5. 战略决策（人类决定）

- [2026-07-04] "method necessity"转为"evaluation/baseline-gated study"是不可接受的。
- [2026-07-04] 不要放松 0.88 gate, 而是尝试一次 locked-protocol coverage 修复。同意。
- [2026-07-04] 产生 hard cells 的关键：需要空间局部化的 residual failure、偏向高风险边界的 objective 分布、参数化 stress-test suite。同意。

6. 下一步行动

§5 三条都已落实到实验设计：没有降级 method necessity，没有放松 0.88 gate，R25 通过空间局部化 residual failure + 高风险 boundary objectives 产生 30 个 hard cells。下一轮不是改 claim，而是把当前正信号变得更可审计、更可复现、更少 post-hoc。

6.1 优先级

1. P0: R27 同步并校验 R25 cells/decisions 与 R26 decisions，修正 R26 aggregate 20/30 vs triple-strict 16/30 的 provenance。
2. P0: R28 对 10 个 boundary/gamma-ceiling failure 做 scale-estimator rescue/诊断。
3. P0: R29 用 calibration-only 规则选择 k/q，再在 held-out objectives 上评估，解决 post-hoc frontier 质疑。
4. P1: R30 独立复核 R25 hard cells 的 coverage cliff，确认哪些 cell 在更大 random-test 下仍过 0.88 gate。

Agent 执行层

A0. Audit Response

audit issue	scientist response	action/evidence	status
AUD-CRIT-001 R26 20/30 count inflated	Accept. The claim-bearing count is 16/30 triple-strict, not the report's two separate 20-count aggregates.	§4.1/§4.2 use 16/30; R27 will write a correction/provenance receipt so future agents do not reuse the stale manifest summary.	planned
AUD-CRIT-002 R25/R26 raw evidence remote-only	Accept. Aggregate reports are not enough for the next audit.	R27 syncs or hashes R25 10368 cells + 10368 decisions and R26 1800 decisions; local counts currently 0/0/0.	planned
AUD-MAJOR-001 boundary/gamma-ceiling failure	Accept. This is a structural limitation.	§4.3 records it; R28 tests boundary-aware scale estimators and extended gamma diagnostics on the 10 failing cells.	planned
AUD-MAJOR-002 kNN hyperparameter sensitivity	Accept. R24 has only 1/20 configs at <=5% cost; R26 used per-cell post-hoc best.	R29 chooses k/q using calibration-only UCB+cost tie-break before held-out evaluation.	planned
AUD-MAJOR-003 R25 coverage concentration	Accept. 26/30 hard cells have random_cov_min in [0.881,0.889].	R30 reruns independent larger random-test coverage and reports which cells remain valid under the fixed 0.88 gate.	planned
AUD-MAJOR-004 R24 gap below 0.08	Accept. Williams/sym is valid by violation but not by gap gate.	§4.1 frames R24 as real-case support plus sensitivity warning, not primary breadth evidence.	resolved
AUD-MAJOR-005 stale A0/A1/§6	Accept.	This STATE replaces completed R23-R26 plans with R27-R30 and removes collected rows from A3.	resolved

A1. Experiments-to-do

Task Group A: R25/R26 evidence sync and correction

- `can_split: false` ; `depends_on: none` ; `priority: P0` ; `runs: R27` .
- Why first: R28-R30 should not build on remote-only raw evidence.

Run: R27-r25-r26-cell-sync-validation

- Task Group: A
- Advances: auditability of §4 R25/R26 claims; no new scientific claim.
- Config: sync or remote-hash `results/r25_parameterized_localized_stress_suite_v2/{cells,decisions}` and `results/r26_scale_aware_on_r25_hard_cells/decisions` ; recompute R26 triple-strict count from `per_cell_best.csv` .
- Input assets: R25/R26 manifests and remote paths on `node05:/home/user3/youran/ocl-raghav` .
- Expected outputs: `results/r27_r25_r26_cell_sync_validation/manifest.json` , counts, SHA256 directory digests, mismatch report, and a correction note that claim-bearing R26 count is 16/30 triple-strict.
- Priority: MUST-RUN; Compute cost: <2 CPU-h plus transfer; Server: node05 + local-cpu; remote_dir: `/home/user3/youran/ocl-raghav` .
- Success criterion: R25 has 10368 cell JSON + 10368 decision JSONL verifiable locally or by digest; R26 has 1800 decision JSONL verifiable; recomputed counts match §4.
- Failure interpretation: if files are missing or hashes mismatch, R25/R26 claims remain aggregate-only and cannot be used as main evidence.
- Evaluation type: provenance/sync validation; Claim ceiling: operational receipt only.

Task Group B: boundary failure rescue

- `can_split: true` ; `depends_on: R27` ; `priority: P0` ; `runs: R28` .

Run: R28-boundary-gamma-ceiling-rescue

- Task Group: B
- Advances: explains the 10 R26 all-seeds failures and tests whether the method limitation is fixable without changing gates.
- Config: on the 10 failing R26 cells, compare current kNN to boundary-aware scale features: anisotropic/Mahalanobis kNN, distance-to-boundary bins, local residual quantile by pocket-distance, and extended gamma grid `[0,0.25,0.5,1,1.5,2,3,5,7.5,10,15,20,30]` .
- Input assets: R25 hard-cell definitions, R26 failing cell list, same train/cal/objective seeds as R26.
- Expected outputs: `results/r28_boundary_gamma_ceiling_rescue/manifest.json` , per-cell frontier, gamma-ceiling diagnostic, failure-mode table, and decision JSONL.
- Priority: MUST-RUN; Compute cost: ~20-40 CPU-h; Server: node05; remote_dir: `/home/user3/youran/ocl-raghav` .
- Success criterion: at least 6/10 failing cells become all-seeds-pass with violation<=0.10 and cost<=5%, or diagnostics show the required margin exceeds 5% cost for structural reasons.
- Failure interpretation: if all variants fail cheaply, boundary residuals need a different scale estimator or active data design, not claim relaxation.
- Evaluation type: method ablation/diagnosis; Claim ceiling: strengthens limitation/rescue story, not a new primary claim alone.

Task Group C: pre-registered k/q selection

- `can_split: true` ; `depends_on: R27` ; `priority: P0` ; `runs: R29` .

Run: R29-knn-kq-selection-protocol

- Task Group: C
- Advances: removes post-hoc best-k/q concern from R26.
- Config: for each R25 hard cell, choose k/q on objective-cal only using UCB selected-risk pass first, then minimum matched cost tie-break; evaluate exactly once on held-out objective-test. Compare to R26 post-hoc oracle best.
- Input assets: R25 hard cells, R26 frontier, R26 decision files after R27, same k/q grid `{5,10,15,30} x {0.5,0.65,0.75,0.9,0.95}`.
- Expected outputs: `results/r29_knn_kq_selection_protocol/manifest.json`, selected k/q table, held-out metrics, oracle-best gap, and per-seed pass table.
- Priority: MUST-RUN; Compute cost: ~10-25 CPU-h if reusing R26 decisions, ~60 CPU-h if rerun needed; Server: node05 + local-cpu; remote_dir: `/home/user3/youran/ocl-raghav`.
- Success criterion: calibration-only selection retains at least 12/30 triple-strict cells and median excess matched cost <=2pp versus post-hoc best.
- Failure interpretation: if selection collapses, R26 remains proof of existence but not a deployable protocol; next method must learn/regularize k/q.
- Evaluation type: selection protocol validation; Claim ceiling: required for method evidence to survive hyperparameter-sensitivity review.

Task Group D: coverage-cliff robustness

- `can_split: true`; `depends_on: R27`; `priority: P1`; `runs: R30`.

Run: R30-hard-cell-coverage-robustness

- Task Group: D
- Advances: tests whether R25 hard cells are artifacts of barely passing random coverage.
- Config: rerun random coverage for all 30 R25 hard cells with independent random-test seeds and `n_random_test=5000`; also report stricter post-hoc gates 0.90 and 0.92 without changing the official 0.88 gate.
- Input assets: R25 hard-cell definitions, original surrogate/data seeds, R25 aggregate reports.
- Expected outputs: `results/r30_hard_cell_coverage_robustness/manifest.json`, independent coverage table, binomial CIs, valid-cell subset under 0.88/0.90/0.92, and corner-family failure note.
- Priority: MUST-RUN if R27 verifies raw evidence; Compute cost: ~20-40 CPU-h; Server: node05; remote_dir: `/home/user3/youran/ocl-raghav`.
- Success criterion: at least 10/30 cells, including at least 2 MLP cells, remain coverage-valid at 0.88 under independent sampling.
- Failure interpretation: if few remain valid, generate a larger or better-controlled localized suite rather than relaxing the gate.
- Evaluation type: robustness/sanity; Claim ceiling: validates breadth of R25 hard cells.

A2. 实验详细规格

- Hard-cell gate remains unchanged: random coverage min >=0.88; selected gap >=0.08; static selected violation >=0.15; no simple baseline with selected violation <=0.10 at <=5% matched objective degradation.
- R26 claim-bearing count is 16/30 triple-strict. Do not reuse report field `n_cells_violation_pass_and_cheap=20` as full pass count; it omits `all_seeds_pass`.
- Coverage robustness may add stricter 0.90/0.92 diagnostics, but those are diagnostics only and do not rewrite the official 0.88 gate.
- R29 k/q selection must use only calibration objectives; no test-objective or full-frontier post-hoc choice.
- All matched costs use the both-method finite-and-oracle-feasible objective subset.
- Every run writes canonical `results/<run>/manifest.json`, `launch.sh`, `nohup.log`, `job.pid`, source commit, input asset ids, sync status, and key command/config.
- node05 launch: current health 1h CPU 3%, 12h 12%, disk 10.6T free; use Python 3.12 env if `uv` selects py3.13 and scipy 1.12 would build from source.

A3. Runs

run	server	remote_dir	launched_at	session_id	crash_count	phase
r27-r25-r26-cell-sync-validation	node05 + local-cpu	/home/user3/youran/ocl-raghav	2026-07-08T00:15:43-04:00	local-rsync-r27-0708	0	collected
r28-boundary-gamma-ceiling-rescue	node05	/home/user3/youran/ocl-raghav	2026-07-09T02:15:12-04:00	ocl-raghav-r28-boundary-gamma-0709-resume5	1	running
r29-knn-kq-selection-protocol	node05 + local-cpu	/home/user3/youran/ocl-raghav	2026-07-08T03:20:07-04:00	local-r29-0708-021500	0	collected
r30-hard-cell-coverage-robustness	node05	/home/user3/youran/ocl-raghav	2026-07-08T07:01:54-04:00	ocl-raghav-r30-hardcov-0708-190154	0	collected

A4. 环境

服务器	资源	远程路径	注意
local-cpu	32 threads, 128GB RAM	`/home/youran/AutoResearch/agent-factory-data/workspace/ocl-raghav`	STATE/provenance checks and local sync validation
node05	96 physical cores, 5.9TiB RAM, no GPU	`/home/user3/youran/ocl-raghav`	2026-07-07 23:51 EDT health: 1h CPU util 3%, 12h 12%, disk 10.6T free; use Python 3.12 env

A5. 运行历史

时间	事件
07-09 03:37 EDT	R28 segment5 monitor: elapsed 1h22m, still 1427/1500 in `partial_results.json` , no final report/frontier/failure table. `resume5` remains alive with timeout PID 915590, Python PID 915599, 60 worker children at ~5934% CPU and no traceback; treated as slow CPU-bound batch, not crash.
07-09 05:41 EDT	R28 segment5 direct node05 verify: 1471/1500 (n_done=1471, 1471 decision JSONL), +44 since 03:37. Rate ~12.8/h; 29 remaining est. ~2h16m. Timeout remaining 2h33m — barely fits. Process healthy, 60 workers, no traceback. All remaining tasks are interior_near_boundary MLP non-standard estimators.
07-09 02:15 EDT	R28 segment 5 manually relaunched from the same `results/r28_boundary_gamma_ceiling_rescue/launch.sh` using `tee -a` into `train.log` , screen `ocl-raghav-r28-boundary-gamma-0709-resume5` . Bring-up confirmed `Resume: 1427 tasks already completed` , 73 remaining, timeout PID 915590, Python PID 915599, and 60 worker children at ~5940% CPU. Auto-relaunch root cause: `/tmp/r28_auto_relaunch.sh` used `set -euo pipefail` ; when no orphaned `run_boundary_gamma_ceiling_rescue` process remained after normal timeout, the `remaining=$(ps aux
07-09 14:08 CST	R28 direct node05 verification: `r28-auto-relaunch` and `resume5` are **not** present; only idle `resume4` screen remains after timeout. `partial_results.json` has 1427/1500 tasks and 1427 non-empty decision JSONL files; no `boundary_gamma_rescue_report.json` , `frontier.csv` , or `failure_mode_table.csv` . Remaining 73 tasks are all interior-near-boundary MLP (`w=0.08,a=4.0,rb=2`). Did not relaunch per dispatcher instruction.
07-09 12:02 CST	R28 auto-relaunch configured: screen `r28-auto-relaunch` on node05 waits for seg4 timeout (~12:57 CST), kills orphans, launches seg5 in `ocl-raghav-r28-boundary-gamma-0709-resume5` . Seg4 cost $16.92 (6h × 60 cores × $0.047). Total R28 cost to date: $67.68 (4 segments).
07-09 11:40 CST	R28 segment 4 deep verify: **CORRECTION** — w0p12 (boundary|0.12|1.0|4.0) was actually fully complete (150/150 tasks, 600-row decisions, 0/90000 NaN, feasibility 88.4%); files from segment 3 tail (timestamps 03:44-05:30 CST). Previous 09:28 check misidentified the slow cell. The real slow cell is **interior_near_boundary** (MLP): 60/150 done (standard_knn complete, 3 estimators pending), rate ~10/h. Process alive, ~1h to 6h timeout. **Corrected rescue analysis (cost threshold 0.05 fraction = 5%): 5/10 cells triple-strict** — boundary|0.04|2.0|8.0|mlp (7/50), boundary|0.08|2.0|4.0 (4/50), boundary|0.08|4.0|4.0 (10/50), boundary|0.12|1.0|4.0 aka w0p12 (2/50), interior (15/20). **5/10 cells have 0 viol_pass configs across all estimators** — all share rb8 or high-amplitude boundary/symbolic pattern → structural kNN limitation. Segment 4 cost $16.92.
07-09 09:28 CST	R28 segment 4 (resume4) 2h31m check: 1350/1500 in partial_results, 60 workers at 99% CPU. Misidentified w0p12 as zero-progress (actually done from segment 3).
07-08 18:58 EDT	R28 segment 3 timed out after 6h with 1350/1500 in `partial_results.json` , 1350 nonempty decisions, and no final `boundary_gamma_rescue_report.json` /frontier/failure table. Relaunched segment 4 from the same `launch.sh` in screen `ocl-raghav-r28-boundary-gamma-0708-resume4` using `tee -a` ; bring-up confirmed 150 remaining tasks, timeout PID 2576135, Python PID 2576143, 60 workers at ~100% CPU. Cleaned stale `resume3` screen PIDs 141786/141787. Added segment 3 CPU-equivalent cost +$16.92.
07-08 17:05 EDT	R28 segment 3 direct node05 check: `partial_results.json` has 1326/1500, 1326 nonempty decisions, `rescue_report.json` /frontier/failure table not yet present. Screen/PID 141838 alive, 62 related processes at ~5947% CPU. Timeout still ~18:52 EDT; no relaunch yet.
07-08 14:00 EDT	R28 segment 3 monitor: 1202/1500 at 2h08m. Rate ~~48/h (稳定). 60 workers healthy. ~3.9h left → ~1389/1500 at timeout (~~18:52 EDT); segment 4 ~2.3h for ~111 tasks.
07-08 12:52 EDT	R28 segment 3 resume: segment 2 completed 1102/1500 (timeout as expected). Relaunched in screen `ocl-raghav-r28-boundary-gamma-0708-resume3` PID 141838; 60 workers at 99% CPU; checkpoint resume confirmed "398 remaining". No CPU contention (192 cores, load ~108).
07-08 11:16 EDT	R28 monitor: 956/1500 (63.7%), 573,600 decisions on disk, 60 workers at 99% CPU, no crash. Rate ~150/h → timeout expected ~12:46 EDT; will checkpoint-resume 3rd segment. Verified decision data clean (no NaN in core fields); log `(nans)` from diagnostic fields only.
07-08 09:11 EDT	R28 monitor: node05 screen/PID 2022755 alive, 60 worker processes running at ~5938% CPU; checkpoint `partial_results.json` shows 732/1500 done and 732 nonempty decision JSONL. No traceback; final report/frontier/diagnostic outputs not yet present.
07-08 07:02 EDT	R30 collected: independent `n_random_test=5000` coverage on 30 R25 hard cells; 24/30 remain valid at official 0.88 gate, including 9/9 MLP cells; 0/30 pass diagnostic 0.90/0.92. Wall-clock 7s × 16 CPU workers, cost rounds to $0.00.
07-08 06:46 EDT	R28 pure checkpoint resume: previous 6h timeout left 387/1500 tasks in `partial_results.json` ; relaunched same `launch.sh` on node05 screen `ocl-raghav-r28-boundary-gamma-0708-184631` ; bring-up shows 60 CPU workers and no traceback. Added prior segment CPU-equivalent cost +$16.92.

时间	事件
07-08 06:32 CST	R29 collected: 17/23 triple-strict, 23/23 all-seeds-pass, median excess cost +0.23%. Both success criteria pass. Wall-clock 3.2h local CPU.
07-08 03:20 CST	R29 launched locally (14 CPU workers, n_cal_cost=100 subset); 69 cal tasks (23 cells × 3 seeds), 222 candidate (cell,k,q) combos.
07-08 00:34 EDT	R28 launched on node05 in screen `ocl-raghav-r28-boundary-gamma-0708-003327` ; 10 R26 all-seeds failures, 4 estimator families, 1500 tasks, 60 CPU workers, 6h timeout.
07-08 00:16 EDT	R27 collected: synced R25 cells/decisions and R26 decisions locally, matched node05 directory digests, and recomputed R26 triple-strict count as 16/30.
07-07 23:54 EDT	iter5 audit CRITICAL: R26 full pass count corrected to 16/30 triple-strict; R25/R26 raw evidence remote-only; boundary/gamma ceiling, k/q sensitivity, coverage cliff, and R24 gap caveats require next-round experiments.
07-08 11:52 CST	R26 collected 1800/1800. Report field `n_cells_violation_pass_and_cheap=20` is not triple-strict; actual triple-strict count is 16/30.
07-06 08:04 EDT	R25 collected 10368/10368 cells; 30/216 pass all hard-cell gates; cells/decisions remain remote-only.
07-05 05:31 EDT	R24 collected; random_cov_min=0.881, static violation=0.166, gap=0.058, kNN best violation=0.043 at +3.61%.
07-04 19:50 EDT	R23 collected; R17 cell-level evidence synced and validated.

A6. 已知问题与修复

问题	根因	下一步	状态
zero valid hard cells	R17 0/48 and R21 dominated by α-tight	R25 found 30 hard cells with NO cheap baseline; R26 succeeds on 16/30 triple-strict	resolved
R26 final report crash	`TypeError: must be real number, not NoneType` in `_frontier_rows` line 666 — `.get()` returned None (key existed with null value), not default `float("nan")`	Fixed both occurrences (lines 666, 723) with `is not None` guard; rsynced to node05; reports regenerated from `partial_results.json`	resolved
Williams/sym coverage invalid	baseline-resistant cell had random_cov min 0.859 < 0.88	R24 locked-protocol repair passed coverage and baseline gates; still needs breadth from R25/R26	resolved for Williams/R24
R17 cell-level data remote-only	`sync_status=aggregates_local_cells_remote`	R23 synced/validated cells and decisions	resolved
R25/R26 raw evidence remote-only	aggregate reports were local but R25 cells/decisions and R26 decisions were absent locally	R27 synced raw files, matched node05 digests, and wrote `results/r27_r25_r26_cell_sync_validation/manifest.json`	resolved
kNN boundary failures	10 R26 failures have gamma_mean >=7.17; 9/10 are boundary/sym	R28 1427/1500 done after segment 4 timeout; no segment 5 or auto-relaunch screen is live. Current evidence still shows 5/10 strict rescued, with interpretation-sensitive alternatives of 6/10 if the `cost=None` baseline-infeasible cell counts, and structural failure count 3/10 vs 5/10 depending on strict definition. Remaining 73 tasks are interior-near-boundary MLP non-standard estimators.	MAJOR, R28 incomplete; final report/frontier/failure table absent
R28 auto-relaunch missed segment5	`/tmp/r28_auto_relaunch.sh had set -euo pipefail` ; the orphan-count pipeline returned `grep` exit 1 when no orphaned runner existed, so the script exited before launching `resume5`	Relaunched segment5 manually with the same `launch.sh and tee -a` ; future one-shot auto scripts should use `pgrep -fc ...	
k/q post-hoc sensitivity	R24 has only one <=5% kNN point; R26 uses per-cell best	R29 calibration-only k/q selection: 17/23 triple-strict, 23/23 all-seeds-pass, median excess cost +0.23%	resolved
R25 coverage cliff	26/30 R25 hard cells had original coverage 0.881-0.889	R30 independent 5000-point coverage confirms 24/30 still pass official 0.88 gate, incl. 9/9 MLP; no cells pass diagnostic 0.90/0.92	resolved for official gate; caveat remains

问题	根因	下一步	状态
			for stricter gates

Coder 旁注

R28 segment5 2026-07-09 05:41 EDT 直接 node05 核实：1471/1500（+44 from 1427），1471 decision JSONL 已落盘。速率 ~12.8/h，剩余 29 tasks（全为 interior_near_boundary MLP 非 standard estimator），预计 2h16m 完成。Timeout 还剩 2h33m，刚好够完成。60 workers 健康，无 traceback。
auto-relaunch 根因：`/tmp/r28_auto_relaunch.sh` 在 `set -euo pipefail` 下用 expected-empty `grep|wc` 计数 orphan；无 orphan 时 `grep` 返回 1，脚本在 launch 前退出。
最终 `boundary_gamma_rescue_report.json`、frontier.csv、failure_mode_table.csv 仍未生成，不能标 collected。
现有证据的两种读法请交给 scientist：严格 cost gate 下 5/10 rescued；若 baseline 100% infeasible 导致 `cost=None` 的 `boundary|0.08|4.0|8.0` 也算 rescued，则 6/10。
Structural failure 也有两种读法：只算真实 high-violation cells 是 3/10（viol 0.197-0.470）；把所有未通过 strict gate 的 cell 都算失败则是 5/10。
未创建 cron/定时任务。下次 coder 接手：若已完成→collect evidence 并标 collected；若 timeout 未完成→checkpoint/resume relaunch segment6。

自检清单

1. §5 preserved unchanged: yes.
2. latest audit BLOCKER/CRITICAL/MAJOR answered in A0: yes.
3. Next plan directly addresses every §5 instruction: yes.
4. A1 uses Task Groups with `can_split`, `depends_on`, `priority`, claim/evidence target, inputs, outputs, success criterion, failure interpretation, evaluation type, and claim ceiling: yes.
5. A3 phases use factory states: yes.
6. Collected runs integrated into §4 and removed from A3: yes.

FILES (23)

.claude	4.0 KB · 2026-07-09T05:40:25
.git	4.0 KB · 2026-07-09T05:43:47
.gitignore	0.4 KB · 2026-07-03T15:16:14
.pytest_cache	4.0 KB · 2026-07-03T17:38:18
.ruff_cache	4.0 KB · 2026-07-03T17:07:10
.venv	4.0 KB · 2026-07-03T15:23:44
LESSONS.md	2.9 KB · 2026-07-08T00:06:54
LLM-SR	4.0 KB · 2026-07-03T20:21:56
LLM_OCL_PROJECT_PLAN.md	84.3 KB · 2026-07-02T00:22:19
LLM_SR_Methodology.md	7.7 KB · 2026-07-02T00:22:19
STATE.md	34.8 KB · 2026-07-09T05:43:06
audits	4.0 KB · 2026-07-07T23:56:22
data	4.0 KB · 2026-07-08T06:40:51
experiment-log.md	48.9 KB · 2026-07-09T03:38:02
idea.md	3.7 KB · 2026-07-03T15:05:42
landscape.md	8.4 KB · 2026-07-02T00:31:36
lit-feed.md	1.4 KB · 2026-07-03T15:06:25
proposal.md	23.8 KB · 2026-07-03T00:43:24
pyproject.toml	0.5 KB · 2026-07-03T15:24:55
results	4.0 KB · 2026-07-08T07:01:54
tests	4.0 KB · 2026-07-08T06:57:54
topic.md	2.4 KB · 2026-07-03T15:05:40
uv.lock	24.7 KB · 2026-07-03T15:25:00

LESSONS.MD

Lessons: ocl-raghav

可迁移经验

- [2026-07-03] 当唯一有效 selected-gap cell 同时也是方法失败 cell 时，下一轮必须优先在同一 cell 上扩大安全网格、修复 cost 口径、正面对比强 baseline；新增宽度实验只能作为补证据，不能替代 load-bearing cell 的恢复。
- [2026-07-03] OCL safety-cost frontier 的 objective degradation 必须在 both-method finite-and-oracle-feasible 的 objective 子集上计算。分别在各自 feasible 子集取均值会把 safety cost 与 solver failure 混在一起，可能制造假优势或假劣势。
- [2026-07-03] Scalar additive margins can look safe in aggregate while failing a seed and becoming cost-dominated. Always inspect per-seed margin sweeps and cost frontier before treating selected-risk recovery as claim-bearing.
- [2026-07-04] A selected-gap cell is not automatically method evidence. First run a baseline-gated frontier: if alpha-tightening or a small dose margin closes the selected gap at comparable cost, keep the cell as anti-claim evidence and search for harder cells.
- [2026-07-04] When a broad hard-cell screen returns zero valid cells because alpha/dose closes every large gap, the next search should not just add more generic benchmarks. Keep the gate fixed, parameterize localized residual pockets and boundary-biased objectives, and report the full grid before running method recovery.
- [2026-07-08] Aggregate counters can silently conflate different success sets. For claim-bearing tables, recompute the exact intersection predicate (for example all_seeds_pass AND violation gate AND cost gate) from per-cell rows, and keep raw cell/decision evidence locally auditable or hash-validated.